

System Scope Document

ASC Student Software Hub (AlfredGO)

Group: Kayden Streiff, Justin Denny, Danielle Riewaldt, Mathew Kennedy-Brewer

CISY 7203 Web Programming II

Evan Enke

a. Concise Problem Description

Alfred State provides a central interface for numerous software resources spanning academics, financial services, and campus life, but Alfred State's current Ellucian Experience portal feels incredibly clunky and bloated leaving students struggling to navigate it. Students often struggle to quickly find what they need due to the amount of information presented and lack of clear organization. Additionally, the portal currently does not adequately support users accessing it on mobile devices, which drastically cuts down the portal's effectiveness in staff and students busy schedules where they need quick access to their information on the go.

b. Statement of Problem

The current Ellucian Experience portal presents too much information at once, which creates cognitive overload and makes it difficult for students to find the resource they need. Tools are not clearly organized or prioritized, and navigation is unclear. This forces users to search through multiple areas or click around to find what they need. As a result, students often avoid using the system unless it is required, reducing its overall effectiveness.

As a few examples the current portal does not feature an easily accessible search function, which leaves power users or informed users scanning through several categories hoping they find the resource they need. As it sits right now, the search bar is hidden behind a hamburger menu, even on desktop displays, which leaves a good portion of users ignorant to its existence. The search function also appears to only support key word searches from the titles of each resource card, which could drastically reduce the searches effectiveness if the user isn't ultra specific. The header found on the homepage of the portal is extremely large, leading it to creep into the users screen real estate, which could be increased dramatically. The Ellucian portal does not feature bookmarks or saving particular cards to views, which also affect power users using the portal to access the

same information several times per week. The site also displays several performance and customizability restrictions which reduces the user's motivation to continue using the service.

c. Statements of Proposed Solutions

The proposed solution is a centralized dashboard that improves usability by focusing on simplicity and relevance.

The system will:

- **Prominent search-first design:** A search bar that is auto-focused on page load and supports tag-based + keyword search — directly addressing the buried search issue
- **Bookmark / favorites system:** Allow users to pin frequently accessed resources so repeat tasks require zero scanning
- **Card customization:** Users can adjust card size, rename cards, color-code by category, and combine related resources (e.g., Banner, Degree Works, Canvas in one card)
- **Subcategory filtering:** Nested navigation (e.g., Academics → Classes → Registration) rather than single flat categories
- **Responsive design for both desktop and mobile:** Equally optimized for phone and desktop browsers
- **Role-aware dashboard:** Different default layouts for students vs. staff (
- **Collapsible/expandable cards:** Clicking a card frame opens a detailed view without navigating away

d. Logical Analysis of Proposed System

The system is designed to simplify how users interact with campus tools.

Users will:

1. User accesses the portal and is authenticated via SSO
2. System detects user role and loads appropriate dashboard view
3. Access the dashboard

4. View a list of prioritized resources/tools
5. Quickly identify the resource they need

Or

6. Quickly search for the resource
7. Select the resource/tool
8. Be directed to the appropriate system

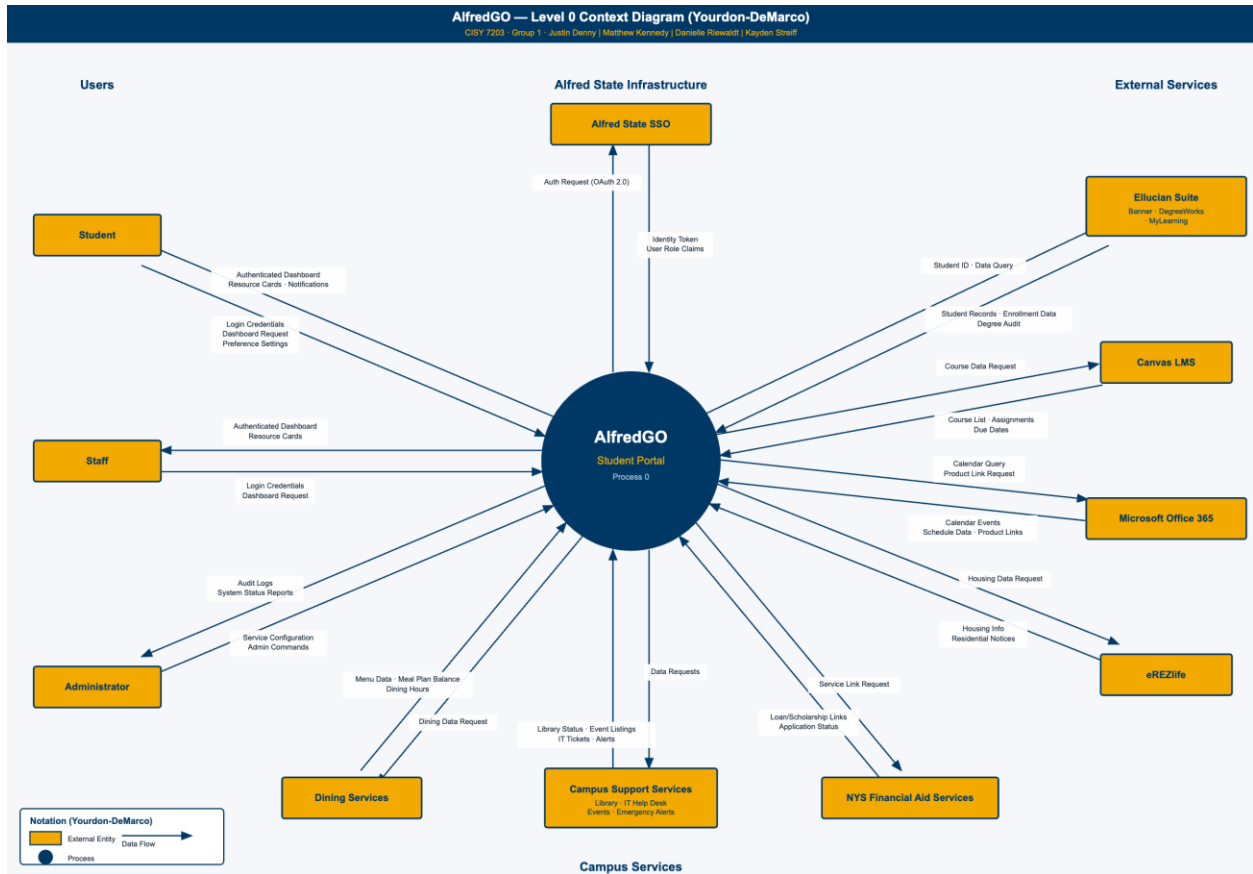
The system removes unnecessary steps and reduces the need for scanning through several vague categories, allowing users to complete tasks more efficiently.

e. Structured Analysis / Logical Design

The system is organized around three main elements:

- Users who access the system
- Tools that provide access to campus services
- Categories that group related tools together

These elements work together to create a simple and organized structure that supports easy navigation and future expansion.



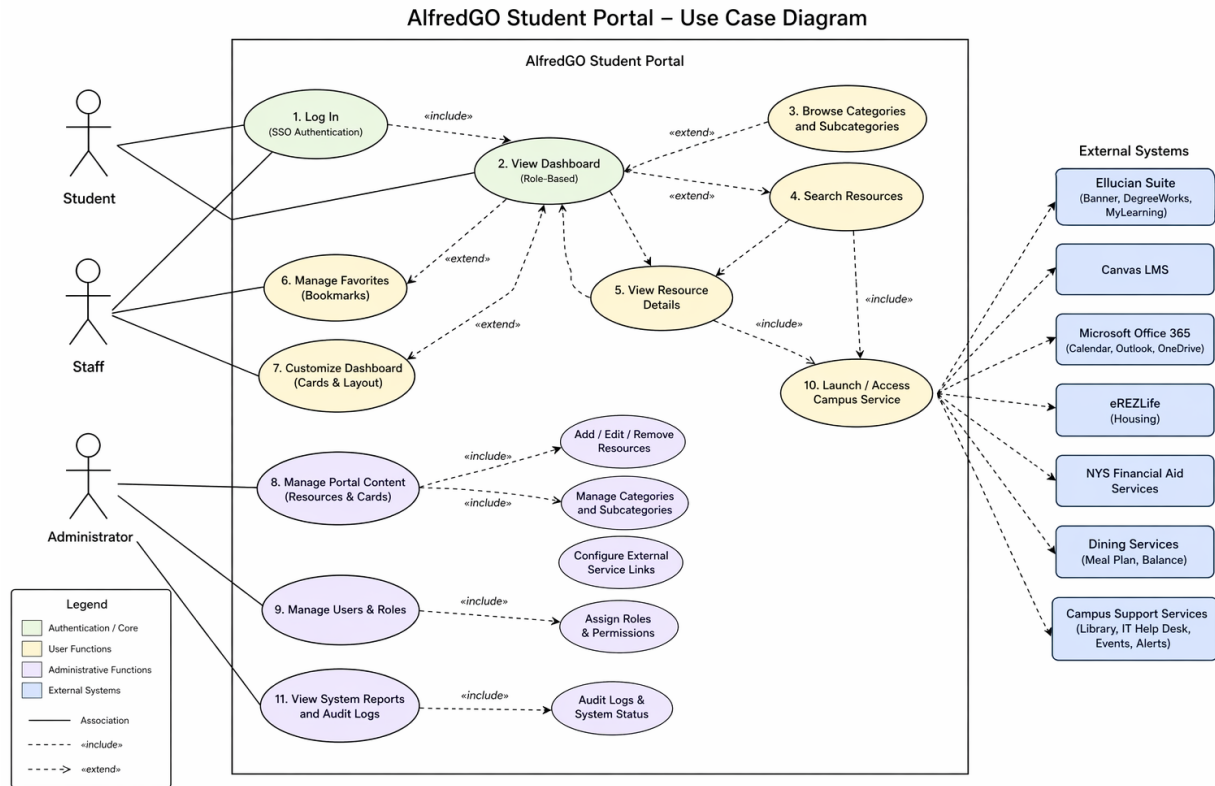
f. Object-Oriented Analysis / Logical Design

The system focuses on interactions between users and tools.

Users interact with the dashboard to:

- View available tools
- Identify relevant options
- Access selected systems

The structure supports clear relationships between users, tools, and categories, ensuring the system remains organized and easy to use.



g. Physical Design

i. User Interface Design

The system will use a dashboard layout that is simple and easy to understand.

The interface will include:

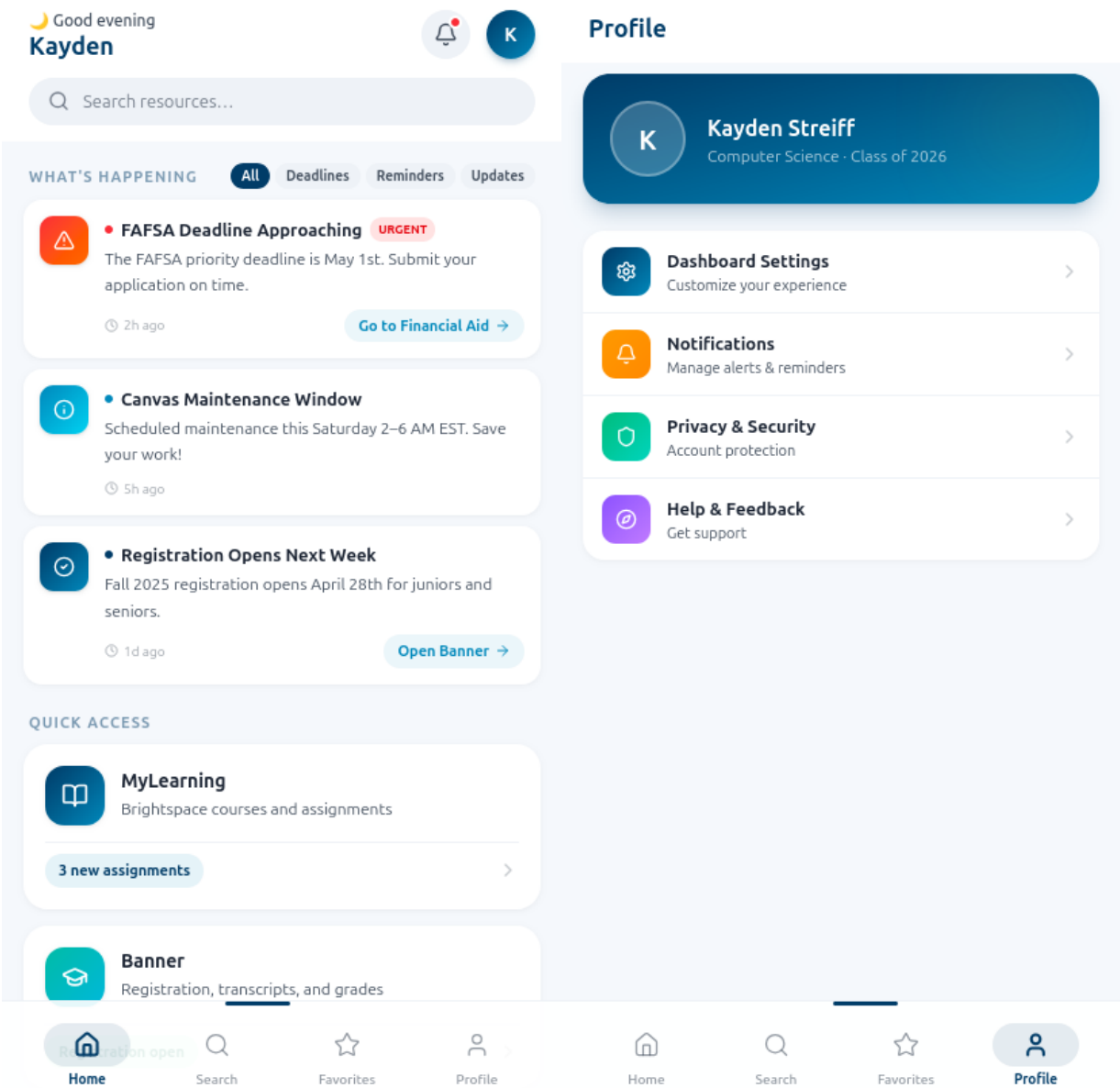
- A main dashboard view
- Clearly organized tool/resource sections
- A layout that highlights the most important tools

Each tool/resource will be presented with:

- A name
- A short description
- A clear way to access it

The design will focus on clarity, simplicity, and ease of use.

Here is a live demo/preview of our design: <https://alfredgo-preview.space.z.ai/>



Your Favorites



Canvas LMS
Courses, assignments, grades





Banner
Registration & transcripts



Search

 Type to search...

TRY SEARCHING FOR

canvas

financial aid

dining

housing

grades



Banner

Registration, transcripts, and student records

CURRENT SCHEDULE

Spring 2025



CS-220 MWF

Data Structures

🕒 9:00–9:50 AM 📍 HETC 101



CS-240 TTh

Web Programming

🕒 11:00 AM–12:15 PM 📍 HETC 205



MATH-230 MWF

Discrete Math

🕒 1:00–1:50 PM 📍 Eng 302



ENG-250 TTh

Tech Writing

🕒 2:00–3:15 PM 📍 Lib 110

QUICK ACTIONS



Registration

Add/drop classes



Grades

View transcripts



Holds

Check account holds



Advisor

Contact info



No Holds



Search



Favorites



Profile

Your account is clear. You are eligible to register for Fall

WHAT'S HAPPENING

All Deadlines Reminders Updates

- 

FAFSA Deadline Approaching URGENT
The FAFSA priority deadline is May 1st. Submit your application on time.
2h ago [Go to Financial Aid](#)
- 

Canvas Maintenance Window
Scheduled maintenance this Saturday 2-6 AM EST. Save your work!
5h ago
- 

Registration Opens Next Week
Fall 2025 registration opens April 28th for Juniors and seniors.
1d ago [Open Banner](#)

QUICK ACCESS

- 

MyLearning
Brightspace courses and assignments
3 new assignments
- 

Banner
Registration, transcripts, and grades
Registration opens
- 

Financial Aid
FAFSA, scholarships, and loans
FAFSA due soon
- 

Email
Outlook, calendar, and OneDrive
2 unread
- 

Degree Works
Degree audit and progress tracking
On track
- 

Dining
Meal plans, balance, and menus
Open until 8PM

RECENT

- 

Canvas LMS
Submitted lab report
2 min ago
- 

Banner
Checked grades
1 hour ago
- 

Outlook Email
Replied to advisor
3 hours ago

ALL RESOURCES

- 

Academics
- 

Canvas LMS
Courses, assignments, grades

Search

Type to search...

TRY SEARCHING FOR

canvas financial aid dining housing grades

K Kayden
AlfredGo

Home

Search

Favorites

Profile

Settings

Your Favorites

Canvas LMS
Courses, assignments, grades



Banner
Registration & transcripts



K Kayden
AlfredGo

Home

Search

Favorites

Profile

Settings

Profile

K

Kayden Streiff

Computer Science - Class of 2026

Dashboard Settings
Customize your experience



Notifications
Manage alerts & reminders

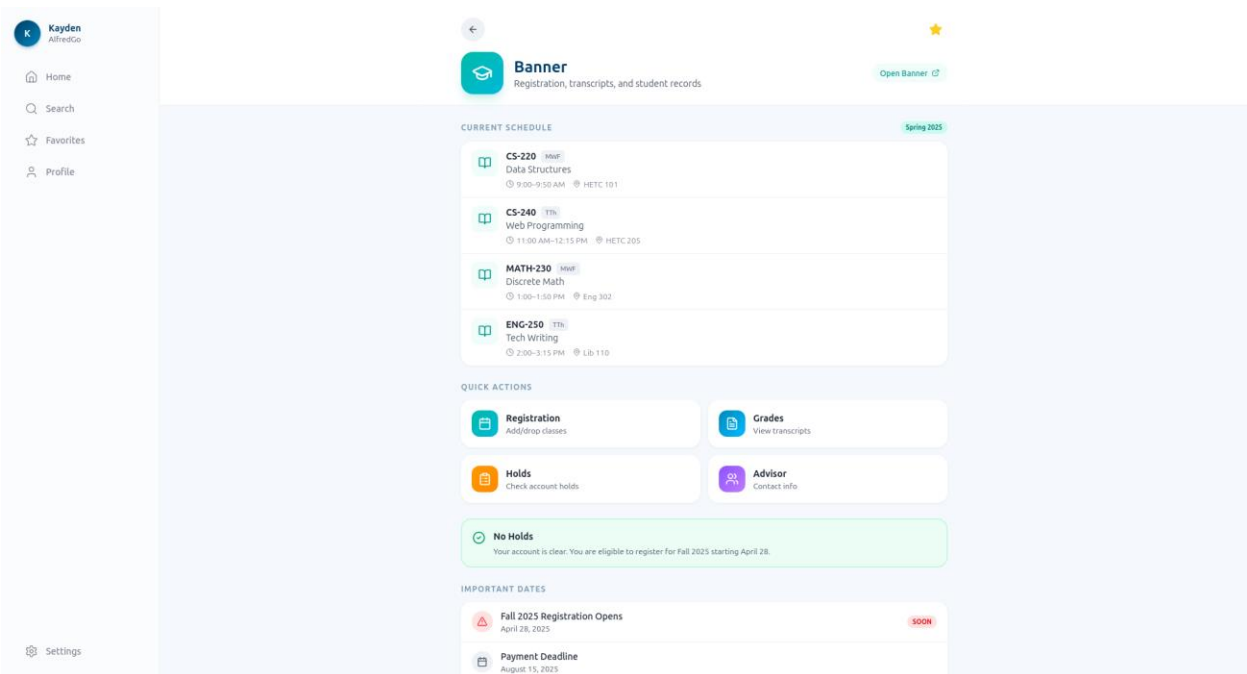


Privacy & Security
Account protection



Help & Feedback
Get support





ii. Data Design

The system will manage information related to:

- Users
- Available tools
- Tool groupings

This information will be structured in a way that supports organization and easy retrieval.

iii. Process Design

The main processes in the system include:

- Displaying tools when the dashboard is accessed
- Allowing users to browse or locate tools
- Redirecting users to the selected system

These processes are designed to minimize effort and reduce the number of steps required to complete tasks.

iv. Physical Storage and Network Configuration

The system will be accessible through a standard web environment and designed to support multiple users accessing it at the same time.

It will be structured in a way that allows for future improvements, such as additional features or expanded functionality.

h. Security Plan

1. Authentication

Alfred State and the Ellucian Portal operate via an SSO (Single Sign on) system, the proposed solution should route through it to maintain the users established flow.

- Alfred State SSO integration (whichever ASC uses — needs confirmation)
- Users authenticate with their existing Alfred State credentials — no new username/password to manage
- Session tokens issued after successful authentication, stored server-side
- Session expiration policy: define timeout period (e.g., 30 minutes of inactivity)
- Token refresh flow for active users

2. Authorization

The proposed system will implement RBAC (Role Based Access Controls), where different access restraints and views will be displayed based on the person currently logged in. For example, an administrator for the college would not need to view registration information, the same way a student does not need access to the college's revenue figures. The solution will define 3 primary roles

- Student — Can view student-relevant tools, customize their dashboard, bookmark tools
- Staff — Can view staff-relevant tools, different default dashboard layout
- Admin — Can manage the tool registry (add/remove/update tools), view usage analytics

The solution should also implement views for unauthorized users, redirecting them back to the homepage or login view.

3. Other Security Concerns

Non-Negotiables (Must Include):

- **FERPA compliance:** Student academic data (grades, schedules) is FERPA-protected. AlfredGO does not store this data — it is fetched live and displayed in-session only.
- **Input validation and sanitization:** Prevent XSS and injection attacks on search inputs and any user-configurable fields
- **HTTPS/TLS:** All traffic should be encrypted in transit for any deployment beyond local development. Production deployment would require a valid SSL certificate and HTTPS enforced on all connections.

Additional security concerns, API rate limiting, dependency auditing or logging and monitoring, fall outside the scope of the current prototype but represent important hardening measures for production deployment.

Including:

- **Dependency security:** Regular audit of third-party packages (npm audit or equivalent)
- **Data minimization:** AlfredGO only stores what it needs (user role, preferences, bookmarks) — it does not replicate sensitive institutional data
- **Rate limiting:** API endpoints protected against abuse
- **Logging and monitoring:** Access logs maintained; anomalies flagged